

# Age/Race Differences in HER2 Testing and in Incidence Rates for Breast Cancer Triple Subtypes

## A Population-Based Study and First Report

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**BACKGROUND:** Although US year 2000 guidelines recommended characterizing breast cancers by human epidermal growth factor receptor 2 (HER2), national cancer registries do not collect HER2, rendering a population-based understanding of HER2 and clinical “triple subtypes” (estrogen receptor [ER] / progesterone receptor [PR] / HER2) largely unknown. We document the population-based prevalence of HER2 testing / status, triple subtypes and present the first report of subtype incidence rates. **METHODS:** Medical records were searched for HER2 on 1842 metropolitan Atlanta females diagnosed with breast cancer during 2003-2004. HER2 testing/status and triple subtypes were analyzed by age, race/ethnicity, tumor factors, socioeconomic status, and treatment. Age-adjusted incidence rates were calculated. **RESULTS:** Over 90% of cases received HER2 testing: 12.6% were positive, 71.7% negative, and 15.7% unknown. HER2 testing compliance was significantly better for women who were younger, of Caucasian or African-American descent, or diagnosed with early stage disease. Incidence rates (per 100,000) were 21.1 for HER2+ tumors and 27.8 for triple-negative tumors, the latter differing by race (36.3 and 19.4 for black and white women, respectively). **CONCLUSIONS:** HER2 recommendations are not uniformly adhered to. Incidence rates for breast cancer triple subtypes differ by age/race. As biologic knowledge is translated into the clinical setting eg, HER2 as a biomarker, it will be incumbent upon national cancer registries to report this information. Incidence rates cautiously extrapolate to an annual burden of 3000 and 17,000 HER2+ tumors for black and white women, respectively, and triple-negative tumors among 5000 and 16,000 respectively. Testing, rate, and burden variations warrant population-based in-depth exploration and clinical translation. *Cancer* 2010;116:2549-59. © 2010 American Cancer Society.

**KEYWORDS:** breast cancer, HER2 testing, triple negative, race, disparity, incidence.

## INTRODUCTION

**Breast** cancer, the most common noncutaneous cancer diagnosed among US females and second most common cause of female cancer death,<sup>1</sup> demonstrates marked diversity in biology and outcome.<sup>2-9</sup> The characterization of breast tumors based on expression of 3 protein biomarkers (estrogen receptor [ER]), progesterone receptor (PR), and human epidermal growth factor receptor 2 (HER2) has been standard recommended care since 2000.<sup>10</sup> In combination, these biomarkers, hereafter referred to as triple subtypes, represent distinct biological and clinical behaviors. However, as of 2009, US

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national cancer registries conducted by the National Cancer Institute's (NCI) Surveillance, Epidemiology, and End Results (SEER) Program or the Centers for Disease Control and Prevention's National Program of Cancer Registries (NPCR) still do not routinely collect or report HER2.<sup>11,12</sup> Thus, the population-based prevalence of HER2 testing, HER2 status, and associated triple subtypes of breast cancer remains largely unknown.

Status of ER, PR, and HER2 is key to clinical management. These biomarkers are predictive in that they determine treatment and treatment-associated outcome, and they are prognostic in that they foretell aggressiveness and outcome, regardless of treatment.<sup>10,13-17</sup> Their combined expression also represents surrogates for the four major intrinsic subtypes: Luminal A, Luminal B, HER2 positive, and triple negative (or basal-like).<sup>6</sup> ER negative (ER-) and PR negative (PR-) breast tumors are more aggressive and occur at an earlier age, more commonly among nonwhite US females (particularly black women).<sup>4,18</sup> ER and PR determine prescription of targeted hormonal therapies with unequivocal clinical utility.<sup>13-15</sup> Prior to targeted therapy, HER2+ tumors portended some of the worst prognoses<sup>19</sup> but are now benefited by another effective targeted treatment; trastuzumab (Herceptin; Genentech, San Francisco, Calif)<sup>13</sup> is currently recommended for early stage, as well as advanced, breast cancer.<sup>13,16,17</sup> The "triple negative" phenotype (ER-/PR-/HER2-) has recently emerged as a distinct subtype, for which there is a lack of effective targeted treatments.<sup>20-25</sup> We and others have recently shown that compared with white women, triple-negative tumors are more prevalent among younger, black and Hispanic women.<sup>21,22,26</sup>

This study focuses on the three existing key biomarkers used in prognosis and treatment decisions: ER, PR, and HER2. It is crucial to investigate these parameters among age, racial/ethnic, and socioeconomic status (SES) groups, where both adherence to standard clinical practices<sup>27-30</sup> and tumor biology may differ.<sup>31-33</sup> To wit, we investigated the population-based prevalence of HER2 testing, HER2 status, and triple subtypes among women diagnosed with primary invasive breast cancer in 2 large metropolitan Atlanta counties during 2003-2004. We also report the first ever incidence rates for breast cancer subtypes and extrapolate our findings into an annual US burden of new cases for black and white women.

## METHODS

### Population

All cases of primary invasive breast cancer diagnosed among female residents of Fulton and Dekalb counties in

metropolitan Atlanta during 2003-2004 were considered for inclusion, representing approximately 1.6 million residents (46% white, 48% black, and 6% Hispanic/Asian/other).

The study population was identified through the NCI's Metropolitan Atlanta SEER Registry in conjunction with the Georgia Comprehensive Cancer Registry (GCCR). Of 2523 abstracts from 48 reporting facilities, 587 were duplicates, resulting in 1936 unique cases. Of these, we excluded patients with multiple primaries or bilateral disease ( $n = 82$ ) or diagnosed by death certificate ( $n = 12$ ). Thus, the final study population consisted of 1842 women.

### Data Collection/Description

We obtained data from the Atlanta SEER registry and GCCR, which routinely collect information regarding sociodemographics, diagnosis, stage, ER, PR, and first-course treatment, but not HER2. We abstracted HER2 information from medical records or registry abstracts. We also "re-abstracted" ER and PR. (Concordance for SEER ER and PR, compared with the re-abstracted ER and PR, was 98.1%; those that were discordant were re-reviewed. We also captured ER and PR on about 50% of tumors categorized as "unknown" by SEER.)

Facility personnel were consulted by the regional registry coordinator for Fulton/Dekalb counties (JHA) and/or the study principal investigator (MJL). Data were entered onto a secured laptop with a boot password and military-grade encryption software.<sup>34</sup> Completed data were stored on the secured server at the Atlanta SEER registry. Institutional review board approval was obtained from Emory University (where the SEER registry is located) and the Georgia Department of Human Resources (GDHR), the GCCR parent body.

HER2 was categorized as assay performed (yes vs no-unknown), and test results were classified as positive (+), negative (-), or unknown based on immunohistochemical (IHC) or fluorescent in situ hybridization (FISH) assay, according to standard recommendations.<sup>35</sup> IHC values of 0 or 1+ were considered negative, 3+ were considered positive. Gene amplification by FISH assay was necessary to confirm 2+ IHC values.

Race/ethnic groups were categorized as white (non-Hispanic), black (non-Hispanic), Hispanic, and other (primarily Asian). Census-tract based SES was constructed by linking the registry census tract variable, based on the street address of the patient at the time of diagnosis, with 2000 US census data.<sup>36</sup> The census tracts were

**Table 1.** HER2 Assays and Results Among Women Diagnosed With Primary Invasive Breast Cancer, Atlanta, GA, 2003-2004

| Characteristic     | HER2 Assay Performed |        |             |        |       | HER2 Result <sup>d,e</sup> |        |                    |        |                  |        |                     |
|--------------------|----------------------|--------|-------------|--------|-------|----------------------------|--------|--------------------|--------|------------------|--------|---------------------|
|                    | Yes<br>n=1660        |        | No<br>n=182 |        | P     | Positive<br>n=232          |        | Negative<br>n=1320 |        | Unknown<br>n=290 |        | P                   |
|                    | No.                  | %      | No.         | %      |       | No.                        | %      | No.                | %      | No.              | %      |                     |
| Race/ethnicity     |                      |        |             |        | .0461 |                            |        |                    |        |                  |        | .0004               |
| White              | 868                  | 89.8   | 99          | 10.2   |       | 108                        | 11.2   | 713                | 73.7   | 146              | 15.1   | .0076 <sup>a</sup>  |
| Black              | 743                  | 91.3   | 71          | 8.7    |       | 110                        | 13.5   | 579                | 71.1   | 125              | 15.4   |                     |
| Hispanic           | 23                   | 79.3   | 6           | 20.7   |       | 8                          | 27.6   | 13                 | 44.8   | 8                | 27.6   |                     |
| Other <sup>b</sup> | 26                   | 81.3   | 6           | 18.8   |       | 6                          | 18.8   | 15                 | 46.9   | 11               | 34.4   |                     |
| Age                |                      |        |             |        | .0771 |                            |        |                    |        |                  |        | <.0001              |
| <40                | 127                  | 93.4   | 9           | 6.6    |       | 27                         | 19.9   | 86                 | 63.2   | 23               | 16.9   | <.0001 <sup>a</sup> |
| 40-49              | 326                  | 89.8   | 37          | 10.2   |       | 62                         | 17.1   | 244                | 67.2   | 57               | 15.7   |                     |
| 50-59              | 470                  | 91.3   | 45          | 8.7    |       | 69                         | 13.4   | 374                | 72.6   | 72               | 14.0   |                     |
| 60-69              | 324                  | 90.0   | 36          | 10.0   |       | 32                         | 8.9    | 270                | 75.0   | 58               | 16.1   |                     |
| 70-79              | 252                  | 91.0   | 25          | 9.0    |       | 29                         | 10.5   | 206                | 74.4   | 42               | 15.2   |                     |
| ≥80                | 161                  | 84.3   | 30          | 15.7   |       | 8                          | 4.2    | 143                | 74.9   | 40               | 20.9   |                     |
| Mean age (SD)      | 58.76                | (14.1) | 61.28       | (15.5) | .024  | 54.64                      | (13.8) | 59.58              | (14.0) | 59.92            | (15.5) | <.0001 <sup>a</sup> |
| SES <sup>c</sup>   |                      |        |             |        | .9565 |                            |        |                    |        |                  |        | .8229               |
| <5%                | 548                  | 90.3   | 59          | 9.7    |       | 73                         | 12.0   | 447                | 73.6   | 87               | 14.3   | .6222 <sup>a</sup>  |
| 5<10%              | 435                  | 90.8   | 44          | 9.2    |       | 61                         | 12.7   | 341                | 71.2   | 77               | 16.1   |                     |
| 10<20%             | 390                  | 90.1   | 43          | 9.9    |       | 62                         | 14.3   | 298                | 68.8   | 73               | 16.9   |                     |
| ≥20%               | 286                  | 89.7   | 33          | 10.3   |       | 36                         | 11.3   | 233                | 73.0   | 50               | 15.7   |                     |
| Unknown            | 1                    | 25.0   | 3           | 75.0   |       | 0                          | 0.0    | 1                  | 25.0   | 3                | 75.0   |                     |
| AJCC Stage         |                      |        |             |        | .002  |                            |        |                    |        |                  |        | <.0001              |
| I                  | 662                  | 92.2   | 56          | 7.8    |       | 79                         | 11.0   | 539                | 75.1   | 100              | 13.9   | .4135 <sup>a</sup>  |
| II                 | 628                  | 93.5   | 44          | 6.6    |       | 83                         | 12.4   | 500                | 74.4   | 89               | 13.2   |                     |
| III                | 158                  | 94.1   | 10          | 6.0    |       | 22                         | 13.1   | 109                | 64.9   | 37               | 22.0   |                     |
| IV                 | 75                   | 80.7   | 18          | 9.9    |       | 12                         | 12.9   | 55                 | 59.1   | 26               | 28.0   |                     |
| Unknown            | 137                  | 71.7   | 54          | 29.7   |       | 16                         | 8.4    | 117                | 61.3   | 58               | 30.4   |                     |

HER2 indicates human epidermal growth factor receptor 2; SD, standard deviation; AJCC, American Joint Committee on Cancer; NOS, not otherwise stated; SES, socioeconomic status; IHC, immunohistochemistry; FISH, fluorescent in situ hybridization; -, not applicable

<sup>a</sup>P-values with unknowns deleted.

<sup>b</sup>>80.0% Asian.

<sup>c</sup>% living below the federally defined poverty level.

<sup>d</sup>FISH assay performed among those w 2+ IHC positive tumors, See Figure 1

<sup>e</sup>5 additional patients were positive by FISH, but results of IHC could not be found.

aggregated into four categories, which represent the percent of the census tract population living below the poverty level.<sup>37,38</sup> Treatment groups were dichotomized into “yes/no”; the “no” group included those women for whom treatment was contraindicated, recommended but not received, or unknown.

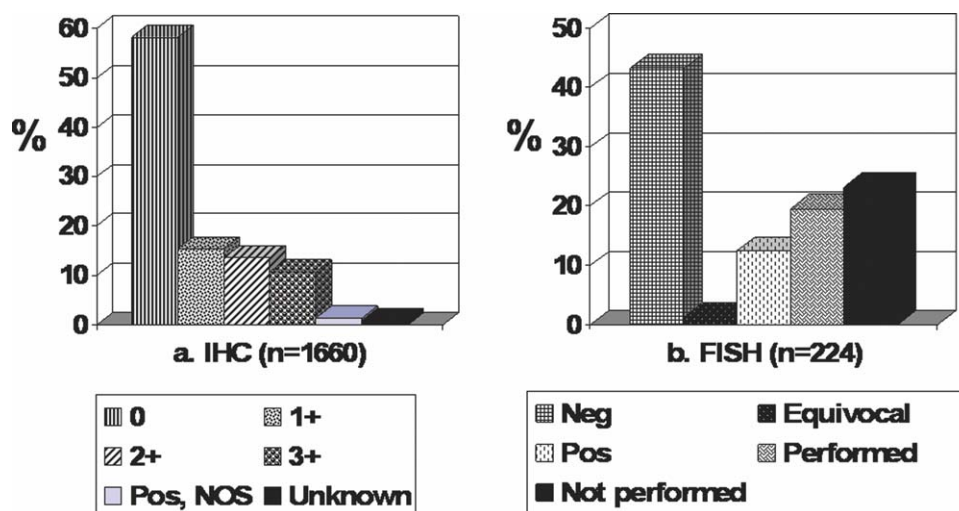
### Data Analyses

Patients were analyzed by HER2 testing status, HER2 results, race/ethnicity, and triple subtypes. Frequency distributions and  $\chi^2$  tests of independence were calculated. Triple subtypes were based on whether expression of ER, PR, and HER2 was + or -: ER-/PR-/HER2- (triple negative), ER-/PR-/HER2+, ER+/PR+/HER2-, and ER+/PR+/HER2+. The subtypes were analyzed by race/ethnicity and other patient-related, tumor-related, and

treatment-related factors. Adjusted odds ratios (ORs) and 95% confidence intervals (CIs) were calculated. Multivariate model variables were determined a priori, based on literature review. Analyses were conducted using SAS version 9.1 (Cary, NC), statistical significance  $P<.05$ . Age adjusted incidence rates were calculated based on the 2000 US census data representing our population, using SEER\*prep and SEER\*stat.

### RESULTS

Of the 1842 women in our study, evidence of HER2 testing was found for 90.1% and did not differ by SES (Table 1). Tumors of older women, Hispanics and other race/ethnicities, and stage IV and unknown stage disease were less likely to be tested.



**Figure 1.** HER2 assay percentages by (a) IHC and (b) FISH are depicted. IHC indicates immunohistochemistry; FISH, fluorescent in situ hybridization.

HER2+ breast cancers represented 12.6% of the cases, 71.7% were HER2– and 15.7% were indeterminable or not assayed (Table 1). Younger women, Hispanics, and other race/ethnicities were most likely to be diagnosed with HER2+ tumors, but differences by SES were not noted.

Tumors scored by IHC most often stained as 0 (58.3%) or 1+ (15.4%), (Fig. 1a). Of the 13.5% scored as 2+, confirmation by FISH assay was documented for nearly 77%; over 43% were negative, 12.5% were positive, and 1.3% equivocal (Fig. 1b). For 19.6% of the 2+ IHC cases, FISH assays were ordered, but results were not found. Of the 232 HER2+ tumors, 178 were 3+IHC, 28 were 2+IHC/FISH+, and 5 were FISH+ with no record of IHC testing (data not shown).

Table 2 displays characteristics by race/ethnicity restricted to black and white women. (The study population was 52.5% white, 44.2% black, 1.6% Hispanic, and 1.7% other race/ethnicity, but meaningful study of Hispanic or other race/ethnicity women was prohibited by limited sample sizes). Black women were significantly more likely than white women to be diagnosed at earlier age, later stage and with tumors that were high grade, and ER and/or PR negative. HER2+ tumors did not differ by race/ethnicity. Triple-negative tumors were most prevalent among black women, accounting for 22.6% of their tumors (and 17.2% of Hispanic women; data not shown), in contrast to 10.4% for white women.

ER, PR, and HER2 information was complete for 1501 women (84.3%) and did not differ by race. The most

prevalent triple subtype was ER+/PR+/HER2– (66.6%), followed by triple-negative tumors (19.0%) (Table 3). HER2+ tumors were least prevalent and fairly equally divided among ER–PR–HER2+ (6.0%) and ER+/PR+/HER2+ (8.5%) subtypes, but HER2+ prevalence was much higher among women aged <50 years (20%) than women aged ≥50 years (12%; data not shown). triple-negative tumors were more prevalent among black women (26.8%) than white women (12.4%) and associated with women who were younger and diagnosed with later stage and larger or higher grade tumors. Triple-negative tumors comprised 28.1% and 14.8% of cancers in black and white women aged <50 years and of those aged ≥50 years 26.2% and 11.7%, respectively (data not shown).

Patients with ER–PR– subtypes (including triple negative or ER–/PR–/HER2+) were most likely to receive chemotherapy (Table 3). Of those with HER2+ disease, women with ER–/PR– disease were more likely to receive Trastuzumab than those with positive ER/PR status.

With the ER+/PR+/HER2– subtype serving as referent in multivariate logistic analyses, triple-negative tumors were independently associated with black race/ethnicity (odds ratio [OR], 2.42), the highest poverty level (OR, 2.35), stage IV disease (OR, 2.35), and tumors of intermediate or high grade, but not age (Table 4). Black women were also nonstatistically more likely to be diagnosed with HER2+ tumors.

Age-adjusted incidence rates differed by race and age (Table 5). Incidence (per 100,000) was highest for ER+/

**Table 2.** Characteristics of Women Diagnosed With Breast Cancer by Race/Ethnicity,<sup>a</sup> Atlanta, GA, 2003-2004

| Characteristic         | White<br>(n= 967) |      | Black<br>(n=814) |        | P-Value <sup>b</sup> |
|------------------------|-------------------|------|------------------|--------|----------------------|
|                        | No.               | %    | No.              | %      |                      |
| <b>Age</b>             |                   |      |                  |        | <.001                |
| <40                    | 58                | 6.0  | 67               | 8.2    |                      |
| 40-49                  | 152               | 15.7 | 191              | 23.5   |                      |
| 50-59                  | 266               | 27.5 | 239              | 29.4   |                      |
| 60-69                  | 200               | 20.7 | 152              | 18.7   |                      |
| 70-79                  | 153               | 15.8 | 115              | 14.1   |                      |
| ≥80                    | 138               | 14.3 | 50               | 6.1    |                      |
| Mean age (SD)          | (61.2)            | 14.6 | 56.9             | (13.4) |                      |
| <b>SES<sup>c</sup></b> |                   |      |                  |        | <.001                |
| <5%                    | 459               | 47.5 | 125              | 15.4   |                      |
| 5<10%                  | 261               | 27.0 | 204              | 25.1   |                      |
| 10<20%                 | 204               | 21.1 | 208              | 25.6   |                      |
| ≥20%                   | 43                | 4.5  | 273              | 33.5   |                      |
| Unknown                | 0                 | 0.0  | 4                | 0.5    |                      |
| <b>AJCC Stage</b>      |                   |      |                  |        | <.001                |
| I                      | 456               | 47.2 | 244              | 30.0   |                      |
| II                     | 300               | 31.0 | 347              | 42.6   |                      |
| III                    | 66                | 6.8  | 96               | 11.8   |                      |
| IV                     | 26                | 2.7  | 63               | 7.7    |                      |
| Unknown                | 119               | 12.3 | 64               | 7.9    |                      |
| <b>Tumor size (cm)</b> |                   |      |                  |        | <.001                |
| T1a,b (0-1.0)          | 275               | 28.4 | 129              | 15.9   |                      |
| T1c (1.1-2.0)          | 349               | 36.1 | 220              | 27.0   |                      |
| T2 (2.1-5.0)           | 221               | 22.9 | 274              | 33.7   |                      |
| T3 (>5.0)              | 35                | 3.6  | 78               | 9.6    |                      |
| T4                     | 17                | 1.8  | 45               | 5.5    |                      |
| Unknown                | 70                | 7.2  | 68               | 7.2    |                      |
| <b>Nodal status</b>    |                   |      |                  |        | <.001                |
| Negative               | 617               | 63.8 | 451              | 55.4   |                      |
| Positive               | 301               | 31.1 | 323              | 39.7   |                      |
| Unknown                | 49                | 5.1  | 40               | 4.9    |                      |
| <b>Tumor grade</b>     |                   |      |                  |        | <.001                |
| Low                    | 220               | 22.8 | 125              | 15.4   |                      |
| Intermediate           | 374               | 38.7 | 275              | 33.8   |                      |
| High                   | 273               | 28.2 | 328              | 40.3   |                      |
| Unknown                | 100               | 10.3 | 86               | 10.6   |                      |
| <b>ER status</b>       |                   |      |                  |        | <.001                |
| Positive               | 732               | 75.7 | 506              | 62.2   |                      |
| Negative               | 171               | 17.7 | 267              | 32.8   |                      |
| Unknown                | 64                | 6.6  | 41               | 5.0    |                      |
| <b>PR status</b>       |                   |      |                  |        | <.001                |
| Positive               | 636               | 65.8 | 426              | 52.3   |                      |
| Negative               | 265               | 27.4 | 345              | 42.4   |                      |
| Unknown                | 66                | 6.8  | 43               | 5.3    |                      |
| <b>HER2</b>            |                   |      |                  |        | .574                 |
| Positive               | 108               | 11.2 | 110              | 13.5   |                      |
| Negative               | 713               | 73.7 | 579              | 71.1   |                      |
| Unknown                | 146               | 15.1 | 125              | 15.4   |                      |
| <b>ER/PR</b>           |                   |      |                  |        | <.001                |
| ER+/PR+                | 623               | 64.4 | 416              | 51.1   |                      |
| ER+PR-                 | 109               | 11.3 | 91               | 11.2   |                      |

(Continued)

**Table 2.** (Continued)

| Characteristic         | White<br>(n= 967) |      | Black<br>(n=814) |      | P-Value <sup>b</sup> |
|------------------------|-------------------|------|------------------|------|----------------------|
|                        | No.               | %    | No.              | %    |                      |
| ER-PR+                 | 15                | 1.6  | 13               | 1.6  |                      |
| ER-PR-                 | 156               | 16.1 | 254              | 31.2 |                      |
| Unknown                |                   |      |                  |      |                      |
| <b>Triple subtypes</b> |                   |      |                  |      | <.001                |
| ER-PR-HER2-            | 101               | 10.4 | 184              | 22.6 |                      |
| ER-/PR-/HER2+          | 41                | 4.2  | 49               | 6.0  |                      |
| ER+/PR+/HER2+          | 67                | 6.9  | 60               | 7.4  |                      |
| ER+/PR+/HER2-          | 605               | 62.6 | 394              | 48.4 |                      |
| Unknown                | 153               | 15.8 | 127              | 15.6 |                      |

HER2, human epidermal growth factor receptor 2; SD, standard deviation, AJCC, American Joint Committee on Cancer; NOS, not otherwise stated; SES, socioeconomic status.

<sup>a</sup> Excludes Hispanics (n=29) and other (n=32) race/ethnicity.

<sup>b</sup> P-values with and with unknowns deleted were similar or equal.

<sup>c</sup> % living below the federally defined poverty level.

PR+/HER2- tumors (98.9) but was lower for black than for white women (86.7 and 110.7, respectively). In contrast, triple-negative tumor incidence was 27.8 overall and was higher among black women (36.3) than white women (19.4). The differences persisted when subdivided into ages <50 years versus ages ≥50 years. HER2+ incidence (regardless of ER/PR status) was 21.1 and did not differ by race/ethnicity.

## DISCUSSION

To our knowledge, this is the first study to document HER2 testing and corresponding results in a population-based cancer registry setting and by race/ethnicity/age groups. This is also the first to present subtype incidence rates by race/age groups. Over 90% of patients diagnosed during 2003-2004 received HER2 testing. In accordance with HER2 guidelines,<sup>10</sup> our study indicates that comprehensive capture of HER2 by cancer registries is achievable and practical. The majority of HER2 tumors assayed as 2+ were confirmed by FISH analysis, as is standard recommendation.<sup>35,39</sup> However, we encountered difficulty finding documentation of the actual FISH results; those 96 (43%) IHC 2+ cases lacking subsequent confirmation by FISH were excluded from our multivariate analyses. We did, however, perform sensitivity analyses, assuming all 96 cases were either 1) positive or 2) negative (the more likely scenario). Neither assumption substantially altered our reported results.



**Table 3.** Characteristics of Black and White Women Diagnosed With Breast Cancer by Triple Subtypes, Atlanta, GA, 2003-2004

| Characteristic           | Triple Negative<br>n=285 |        | ER-/PR-/<br>HER2+<br>n=90 |        | ER+/PR+/<br>HER2+<br>n=127 |        | ER+/PR+/<br>HER2-<br>n=999 |        | P      |
|--------------------------|--------------------------|--------|---------------------------|--------|----------------------------|--------|----------------------------|--------|--------|
|                          | No.                      | %      | No.                       | %      | No.                        | %      | No.                        | %      |        |
| <b>Race/ethnicity</b>    |                          |        |                           |        |                            |        |                            |        | <.0001 |
| White                    | 101                      | 12.4   | 41                        | 5.0    | 67                         | 8.2    | 605                        | 74.3   |        |
| Black                    | 184                      | 26.8   | 49                        | 7.1    | 60                         | 8.7    | 394                        | 57.4   |        |
| <b>Age</b>               |                          |        |                           |        |                            |        |                            |        | <.0001 |
| <40                      | 33                       | 32.4   | 13                        | 12.8   | 10                         | 9.8    | 46                         | 45.1   |        |
| 40-49                    | 53                       | 18.3   | 17                        | 5.9    | 40                         | 13.8   | 180                        | 62.1   |        |
| 50-59                    | 91                       | 20.9   | 31                        | 7.1    | 37                         | 8.5    | 277                        | 63.5   |        |
| 60-69                    | 50                       | 16.9   | 11                        | 3.7    | 21                         | 7.1    | 214                        | 72.3   |        |
| 70-79                    | 43                       | 19.1   | 13                        | 5.8    | 14                         | 6.2    | 155                        | 68.9   |        |
| ≥80                      | 15                       | 9.9    | 5                         | 3.3    | 5                          | 3.3    | 127                        | 83.6   |        |
| Mean Age [SD]            | 56.2                     | [14.0] | 54.7                      | [14.4] | 54.6                       | [13.5] | 60.6                       | [13.8] |        |
| <b>SES<sup>a</sup></b>   |                          |        |                           |        |                            |        |                            |        | .2447  |
| <5%                      | 84                       | 16.8   | 27                        | 5.4    | 46                         | 9.2    | 342                        | 68.5   |        |
| 5<10%                    | 75                       | 19.3   | 18                        | 4.6    | 36                         | 9.3    | 260                        | 66.8   |        |
| 10<20%                   | 63                       | 18.1   | 27                        | 7.8    | 29                         | 8.3    | 229                        | 65.8   |        |
| ≥20%                     | 63                       | 29.5   | 18                        | 6.8    | 16                         | 6.1    | 167                        | 63.3   |        |
| Unknown                  | 0                        | 0.0    | 0                         | 0.0    | 0                          | 0.0    | 1                          | 100.0  |        |
| <b>AJCC stage</b>        |                          |        |                           |        |                            |        |                            |        | <.0001 |
| I                        | 84                       | 13.9   | 32                        | 5.3    | 46                         | 7.6    | 443                        | 73.2   |        |
| II                       | 117                      | 20.5   | 33                        | 5.8    | 54                         | 9.5    | 367                        | 64.3   |        |
| III                      | 34                       | 26.4   | 14                        | 10.9   | 9                          | 7.0    | 72                         | 55.8   |        |
| IV                       | 24                       | 36.9   | 6                         | 9.2    | 6                          | 9.2    | 29                         | 44.6   |        |
| Unknown                  | 26                       | 19.8   | 5                         | 3.8    | 12                         | 9.2    | 88                         | 67.2   |        |
| <b>Tumor Size (cm)</b>   |                          |        |                           |        |                            |        |                            |        | <.0001 |
| T1a,b (0-1.0)            | 30                       | 8.6    | 24                        | 6.9    | 22                         | 6.3    | 272                        | 78.2   |        |
| T1c (1.1-2.0)            | 85                       | 16.9   | 21                        | 4.2    | 50                         | 10.0   | 346                        | 68.9   |        |
| T2 (2.1-5.0)             | 102                      | 23.9   | 26                        | 6.1    | 39                         | 9.1    | 260                        | 60.9   |        |
| T3 (>5.0)                | 24                       | 26.7   | 7                         | 7.8    | 5                          | 5.6    | 54                         | 60.0   |        |
| T4                       | 22                       | 46.8   | 6                         | 12.8   | 2                          | 4.3    | 17                         | 36.2   |        |
| Unknown                  | 22                       | 25.3   | 6                         | 6.9    | 9                          | 10.3   | 50                         | 57.5   |        |
| <b>Nodal status</b>      |                          |        |                           |        |                            |        |                            |        | .3625  |
| Negative                 | 174                      | 19.1   | 50                        | 5.5    | 66                         | 7.2    | 622                        | 68.2   |        |
| Positive                 | 101                      | 18.8   | 37                        | 6.9    | 54                         | 10.1   | 344                        | 64.2   |        |
| Unknown                  | 10                       | 18.9   | 3                         | 5.7    | 7                          | 13.2   | 33                         | 62.3   |        |
| <b>Tumor grade</b>       |                          |        |                           |        |                            |        |                            |        | <.0001 |
| Low                      | 10                       | 3.4    | 1                         | 0.3    | 11                         | 3.7    | 272                        | 92.5   |        |
| Intermediate             | 53                       | 9.3    | 19                        | 3.3    | 61                         | 10.7   | 439                        | 76.8   |        |
| High                     | 207                      | 40.5   | 62                        | 12.1   | 49                         | 9.6    | 193                        | 37.8   |        |
| Unknown                  | 15                       | 12.1   | 8                         | 6.5    | 6                          | 4.8    | 95                         | 76.6   |        |
| TREATMENT <sup>b</sup>   |                          |        |                           |        |                            |        |                            |        |        |
| <b>Chemotherapy</b>      |                          |        |                           |        |                            |        |                            |        | <.0001 |
| Yes                      | 173                      | 60.7   | 46                        | 51.1   | 61                         | 48.0   | 367                        | 36.7   |        |
| No                       | 112                      | 39.3   | 44                        | 48.9   | 66                         | 52.0   | 632                        | 63.3   |        |
| <b>Herceptin therapy</b> |                          |        |                           |        |                            |        |                            |        | <.0001 |
| Yes                      | 0                        | 0.0    | 9                         | 10.0   | 3                          | 2.4    | 0                          | 0.0    |        |
| No                       | 285                      | 100.0  | 81                        | 90.0   | 124                        | 97.6   | 999                        | 100.0  |        |
| <b>Hormonal therapy</b>  |                          |        |                           |        |                            |        |                            |        | <.0001 |
| Yes                      | 11                       | 3.9    | 4                         | 4.4    | 32                         | 25.2   | 324                        | 32.4   |        |
| No                       | 274                      | 96.1   | 86                        | 95.6   | 95                         | 74.8   | 675                        | 67.6   |        |

(Continued)

Table 3. (Continued)

| Characteristic   | Triple Negative<br>n=285 |      | ER-/PR-/<br>HER2+<br>n=90 |      | ER+/PR+/<br>HER2+<br>n=127 |      | ER+/PR+/<br>HER2-<br>n=999 |      | P     |
|------------------|--------------------------|------|---------------------------|------|----------------------------|------|----------------------------|------|-------|
|                  | No.                      | %    | No.                       | %    | No.                        | %    | No.                        | %    |       |
| <b>Radiation</b> |                          |      |                           |      |                            |      |                            |      | .1204 |
| Yes              | 118                      | 41.4 | 35                        | 38.9 | 59                         | 46.5 | 479                        | 48.0 |       |
| No               | 167                      | 58.6 | 55                        | 61.1 | 68                         | 53.5 | 520                        | 52.1 |       |
| <b>Surgery</b>   |                          |      |                           |      |                            |      |                            |      | .0031 |
| Yes              | 254                      | 89.1 | 78                        | 86.7 | 112                        | 88.2 | 939                        | 94.0 |       |
| No               | 31                       | 10.9 | 12                        | 13.3 | 15                         | 11.8 | 60                         | 6.0  |       |

ER indicates estrogen receptor; PR, progesterone receptor; HER2, human epidermal growth factor receptor-2; SD, standard deviation; AJCC, American Joint Committee on Cancer; cm (centimeters).

<sup>a</sup> % living below the federally defined poverty level.

<sup>b</sup> Column percentages.

Those whose tumors were not assayed for HER2 were women aged >80 years, of Hispanic or other race/ethnicity, or with advanced disease. For older women, treatment for HER2+ tumors may more often be contraindicated or not feasible. The lower assay prevalence among Hispanic and other race/ethnicities should be investigated further, as our study results could represent inherent sample size bias. Of concern, HER2 assay was less likely among those diagnosed with advanced stage disease, a group that has been demonstrated to derive benefit from HER2-directed therapies if diagnosed with HER2+ tumors.<sup>16</sup>

The prevalence of HER2+ tumors was 15.0% if restricted to those for whom HER2 results were performed and interpretable. This falls within previously reported prevalences, which demonstrated that HER2 gene amplification was present in 15%-30% of invasive breast cancers and is present in most HER2 protein expressing tumors.<sup>40,41</sup> The higher prevalence of HER2+ tumors among Hispanics and other race/ethnicity described herein have not been reported previously. Constrained by limited sample sizes, this merits investigation.

We and others have reported the prevalence of breast cancer subtypes in a population-based setting,<sup>22,26</sup> but we believe that ours is the first study to report *incidence rates*. Triple-negative tumor incidence rates differed markedly. Rates for black women were nearly double those for white women, regardless of age, and at the expense of the more treatable ER+/PR+/HER2- tumors. Regarding prevalence, triple-negative tumors were most often diagnosed among young black women (28.1%), followed by black women aged ≥50 years (26.2%), white women aged <50 years (14.8%), than white women aged ≥50 years (11.7%). In communion and contrast to the prevalences,

triple-negative tumor incidence rates per 100,000 women were higher among black women overall, but highest for those black women aged ≥50 years (69.0) and lowest among white women aged <50 years (8.7). Why whites are particularly prone to the development of ER+/PR+/HER2- breast cancers, especially women aged ≥50 years, remains unknown.<sup>18</sup> This is the major contributing factor to breast cancer rates being higher among white women than black women, but positively contributes to better outcome among white women, as these tumors are the most indolent and remain the most treatable, partly because of the development of targeted therapeutics.<sup>13-15</sup>

Our incidence rates representing 2 large metropolitan Atlanta counties are comparable to the SEER 9 area US rates for the same time period among women aged ≥50 years, but they are higher for women aged <50 years. Thus, the incidence rates we report can be cautiously extrapolated to an annual new burden of US triple-negative cases for 5000 black women and 16,000 white women and HER2+ breast cancers among approximately 3000 and 17,000 respectively.

The prevalence of triple-negative tumors we report is less than that in prior population-based studies, including the Carolina Breast Cancer Study (CBCS) and our Atlanta-based study of women aged <55 years diagnosed in the early 1990s.<sup>22,26</sup> Unlike those studies, which based triple subtype prevalence partly or completely on centralized testing, our current study reflects subtype status as documented in the clinical records from a variety of methodologies and sources. In the CBCS, Carey et al reported that premenopausal black women were most likely to present with triple-negative tumor (39%) versus 14% among premenopausal nonblack women, while equivalent 16% triple-negative tumor prevalences among

**Table 4.** Associations (Adjusted Odds Ratios and 95% Confidence Intervals) for Triple Phenotypes of Breast Cancer, Atlanta, GA, 2003-2004

|                                         | Triple Negative <sup>a,b</sup> |             | ER-/PR-/HER2+ <sup>a,b</sup> |             | ER+/PR+/HER2+ <sup>a,b</sup> |            |
|-----------------------------------------|--------------------------------|-------------|------------------------------|-------------|------------------------------|------------|
|                                         | OR                             | 95% CI      | OR                           | 95% CI      | OR                           | 95% CI     |
| <b>Race/ethnicity</b>                   |                                |             |                              |             |                              |            |
| Black                                   | 2.42                           | 1.69-3.49   | 1.51                         | 0.87-2.62   | 1.35                         | 0.87-2.07  |
| White                                   | 1.00                           | -           | 1.00                         | -           | 1.00                         | -          |
| <b>Age, y</b>                           |                                |             |                              |             |                              |            |
| <40                                     | 1.56                           | 0.86-2.85   | 1.81                         | 0.82-4.00   | 1.38                         | 0.63-3.02  |
| 40-49                                   | 0.64                           | 0.41-1.01   | 0.60                         | 0.31-1.17   | 1.34                         | 0.81-2.21  |
| 50-59                                   | 1.00                           | -           | 1.00                         | -           | 1.00                         | -          |
| 60-69                                   | 0.82                           | 0.52-1.30   | 0.47                         | 0.22-1.00   | 0.76                         | 0.42-1.35  |
| 70-79                                   | 1.09                           | 0.66-1.78   | 0.83                         | 0.39-1.75   | 0.67                         | 0.34-1.31  |
| ≥80                                     | 0.69                           | 0.35-1.34   | 0.51                         | 0.18-1.43   | 0.33                         | 0.12-0.86  |
| <b>Menopausal surrogate<sup>c</sup></b> |                                |             |                              |             |                              |            |
| < 50 years at diagnosis                 | 0.90                           | 0.63-1.27   | 1.12                         | 0.67-1.86   | 1.74                         | 1.16-2.60  |
| +50 years at diagnosis                  | 1.00                           | -           | 1.00                         | -           | 1.00                         | -          |
| <b>SES<sup>d</sup></b>                  |                                |             |                              |             |                              |            |
| ≥20%                                    | 2.35                           | 1.16-4.76   | 1.12                         | 0.51-2.49   | 0.69                         | 0.35-1.37  |
| 5<10%                                   | 1.10                           | 0.63-1.91   | 1.33                         | 0.70-2.55   | 0.91                         | 0.53-1.55  |
| 10<20%                                  | 0.97                           | 0.67-1.40   | 0.97                         | 0.49-1.92   | 1.08                         | 0.66-1.77  |
| <5%                                     | 1.00                           | -           | 1.00                         | -           | 1.00                         | -          |
| <b>AJCC stage<sup>e</sup></b>           |                                |             |                              |             |                              |            |
| IV                                      | 2.35                           | 1.16-4.76   | 1.30                         | 0.45-3.74   | 1.33                         | .050-3.53  |
| III                                     | 1.10                           | 0.63-1.91   | 1.05                         | 0.49-2.25   | 0.73                         | 0.33-1.61  |
| II                                      | 0.97                           | 0.67-1.40   | 0.71                         | 0.40-1.24   | 1.04                         | 0.67-1.62  |
| I                                       | 1.00                           | -           | 1.00                         | -           | 1.00                         | -          |
| <b>Tumor grade</b>                      |                                |             |                              |             |                              |            |
| High                                    | 25.96                          | 13.19-51.12 | 81.32                        | 1.04-598.94 | 5.56                         | 2.76-11.20 |
| Intermediate                            | 3.11                           | 1.54-6.29   | 11.80                        | 1.56-89.19  | 3.33                         | 1.70-6.51  |
| Low                                     | 1.00                           | -           | 1.00                         | -           | 1.00                         | -          |

ER, estrogen receptor; PR, progesterone receptor; HER2, human epidermal growth factor receptor-2; OR, odds ratio; CI, confidence interval; SES, socioeconomic status; AJCC, American Joint Committee on Cancer.

<sup>a</sup>Referent = ER+/PR+/HER2-.

<sup>b</sup>Unknown categories were kept in the logistic models to maintain power but are not presented.

<sup>c</sup>Age at diagnosis is replaced with two age groups representing menopausal surrogates.

<sup>d</sup>% living below the federally defined poverty level.

<sup>e</sup>Replacing AJCC stage with T, N, and M produced similar results.

**Table 5.** Age-Adjusted Incidence Rates<sup>a</sup> of Women Diagnosed With Breast Cancer by Triple Subtypes, Atlanta, GA, 2003-2004

|              | All Breast Cancers | Triple Negative | ER-/PR-/HER2+ | ER+/PR+/HER2+ | ER+/PR+/HER2- |
|--------------|--------------------|-----------------|---------------|---------------|---------------|
| All women    | 176.0              | 27.8            | 8.9           | 12.2          | 98.9          |
| <b>White</b> | 179.3              | 19.4            | 7.5           | 12.6          | 110.7         |
| Age <50      | 66.3               | 8.7             | 3.2           | 7.6           | 37.7          |
| Age 50+      | 357.9              | 36.4            | 14.4          | 20.5          | 226.1         |
| <b>Black</b> | 171.2              | 36.3            | 10.0          | 12.1          | 86.7          |
| Age <50      | 69.6               | 15.7            | 5.7           | 7.4           | 28.1          |
| Age 50+      | 331.8              | 69.0            | 16.9          | 19.5          | 179.2         |

<sup>a</sup>Rates per 100,000 and age-adjusted to the 2000 US population census for Fulton-Dekalb counties.



postmenopausal black and nonblack women were observed. Similarly, in our prior Atlanta-based study, triple-negative tumor prevalence significantly differed between black women and white women, 47% versus 22%, respectively.<sup>26</sup> The 19.0% prevalence of triple-negative tumors we currently report, with the corresponding racial dissimilarities (26.8% black, 17.2% Hispanic, and 12.4%, white), falls between the prior studies and that of a cancer registry study conducted in California.<sup>21</sup> Although missing nearly 45% of the data to construct subtypes, the California study reported a 12.5% triple-negative tumor prevalence, which differed by race (24.6% black, 17.2% Hispanic, and 10.8% white). When we separately analyzed women using an age 50 cutpoint, we did not find marked differences in triple-negative tumor prevalence (22% vs 18%). This contrasts with prior reports of higher triple-negative tumor prevalence among premenopausal women.<sup>22,26</sup> The recent decline in ER+/PR+ tumors among postmenopausal women observed in 2003 and sustained in 2004, making our two age groups more similar, may partially account for the study differences, as could the use of clinical-based, rather than centralized, assay. Despite lack of differences by age groups, our findings that black women were at a 2.42 greater odds of being diagnosed with triple-negative tumors, after adjustment for age, stage, grade, and SES, are concordant with previous population-based studies: the CBCS study (OR, 2.1), our prior Atlanta-based study (OR, 1.9), and a California study (OR, 1.77).<sup>21,22,26</sup> Similar to the California study, we found that living at the lowest poverty level was independently associated with triple-negative tumors, an association not noted in the other tumor subtypes.

Our data indicate that for the most part, treatment based on triple subtypes is concordant with recommended guidelines. More comprehensive investigations, including specific regimens, would help to define treatment patterns and lead to more informed decisions for selection of effective treatments.

We acknowledge strengths and limitations. Triple subtype status was available for only 84% of cases, but this 84% reflects the most complete representation of any population-based study, to date. Also, we compared the populations with versus without subtype information and found no differences in the variables presented. Biomarker status in our study represents a variety of testing methods and sources, rendering them more subject to intercorrelation and intracorrelation error. However, our study more accurately reflects their use in a clinical setting. Representation of Hispanic and other race/ethnicities remains

sparse in metropolitan Atlanta, rendering comprehensive subtype analyses problematic. Without knowledge of epidermal growth factor receptor and CK5-6 expression, some “unclassified” or normal-like tumors as described by Sorlie and Perou et al<sup>6,24</sup> could be misclassified as triple-negative tumors, and we do not know how many. We also did not capture the testing methods for HER2 by IHC (eg, DAKO, Herceptest, CB11, or Homebrew) or by FISH (eg, Vysis Pathyvision or Ventana).<sup>35</sup> Furthermore, population-based registries (including that from which our data were derived) do not collect information on the method of breast cancer detection, ie, screening or symptomatic. Thus, the racial differences we present could, in part, represent differences in access to screening mammography, which can impact stage at diagnosis and other tumor prognostic factors, thereby potentially influencing triple subtype status. However, for our study time period, comparable mammography screening rates for black and white women overall and within age groups have been documented at the national level.<sup>42,43</sup> Also, we have previously reported racial differences in triple subtypes within stage groups,<sup>26</sup> and it is acknowledged that tumors with more aggressive biology (eg, triple-negative tumor) tend to be diagnosed as interval cancers, that is, they are likely to be detected in intervals between recommended screening guidelines.<sup>44</sup> In contrast, nonelderly women (aged <65 years) who lack health insurance (a potential surrogate for those living in impoverished areas) are half as likely to undergo mammography.<sup>42</sup> These studies combined with our current study, which found that race and SES were both independent predictors of subtypes, suggest that race itself is a contributing factor. Finally, generalizing our findings to other populations is limited.

In conclusion, rapid clinical integration of HER2 recommendations in metropolitan Atlanta is widespread, though not equivalently applied. Collecting HER2 information is reasonably achievable. As biologic knowledge is translated into the clinical setting and in that treatment, efficacy is biologically driven, it will be incumbent upon our national and state cancer registries to integrate this knowledge into their reporting schema and to improve upon collection of current standard clinical biomarkers, eg, collection of ER and PR status needs improvement, particularly among state cancer registries that currently do not require their reporting.<sup>45</sup> The assay methods and numeric results should also be captured. HER2 profiles do not differ by SES or black/white race/ethnicity, but black women, and perhaps Hispanic women, are more likely to be diagnosed with triple-negative tumors, which

have no effective targeted therapies and bode a poor prognosis.<sup>40,46</sup> Thus, the subtypes that racial/ethnic groups present with may partially account for differences in outcome.<sup>47</sup> The annual new burden of triple-negative tumor cases of 5000 and 16,000 for black and white women, respectively, a larger burden than HER2+ breast cancers (particularly for black women), exemplifies the need for identifying effective therapies. Clinical trials are currently underway. Ample minority inclusion should be a priority. Our findings, in conjunction with others, demonstrate the critical importance of race/ethnicity and age in ascertaining varying etiologies and outcomes. Sociodemographic factors are clearly implicated in the historically poor outcomes observed for black and other minority women. However, future studies should seek to determine and translate biological mechanisms, with ample study size and minority representation, to adequately assess these relationships and ultimately improve risk and outcome.

## CONFLICT OF INTEREST DISCLOSURES

The authors made no disclosures.

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